Article

CLOP-DiT: Text-Guided Single-Cell Gene Expression Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers

Zeyu Fu [1,*]

[1] State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine, Army Medical University, Chongqing 400038, China; zeyu.fu@tmmu.edu.cn

Abstract

Generating realistic synthetic single-cell gene expression profiles conditioned on natural-language descriptions of cell identity would enable data augmentation, rare cell-type simulation, and hypothesis-driven perturbation studies. We present CLOP-DiT, a two-stage pipeline combining Contrastive Language–Omics Pretraining (CLOP) with a conditional Diffusion Transformer (DiT) trained via flow matching. CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings into a shared 256-dimensional space (3.02M parameters), amplifying inter-type separation from 0.6% to 78% in cosine distance. DiT1D (22.10M parameters, 8 AdaLN-Zero blocks) then performs conditional flow matching in scGPT latent space, and a frozen scGPT decoder maps generated latents to gene expression. Across 100 cell types from 80 datasets (220,304 cells, 1,088 text groups), CLOP-DiT achieves *k*-nearest-neighbor accuracy of 36.9% (37× random chance with 100 classes), steering accuracy of 81.0% (random = 50%), and a diversity ratio of 0.93 under the recommended Midpoint solver at classifier-free guidance scale 1.0. Unconditional generation collapses to random baselines, confirming that text conditioning is the sole driver of type specificity. Downstream biological validation—clustering alignment, classifier transfer (logistic regression accuracy 51.1%), and differential expression concordance (logFC Pearson $r = 0.17$)—demonstrates that generated cells integrate into standard single-cell workflows.

Keywords: single-cell RNA-seq; generative model; diffusion transformer; contrastive learning; text-guided generation; flow matching; gene expression; foundation model

1. Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled the characterization of cellular heterogeneity at unprecedented resolution [1]. However, generating realistic synthetic single-cell expression profiles conditioned on natural-language descriptions of cell identity remains an open challenge. Such a capability would enable data augmentation for rare cell types, hypothesis testing via *in silico* perturbation, and atlas completion without additional sequencing.

Existing approaches to single-cell data generation include variational autoencoders such as scVI [2] and scGen [3], geometric deep generative models [4], and more recently, large-scale single-cell foundation models such as Geneformer [11] that leverage transfer learning for downstream prediction. These methods condition on categorical cell-type labels or perturbation metadata, but none accept free-text biological descriptions as input.

Received:
Revised:
Accepted:
Published:

 https://doi.org/10.3390/biology1010000

In parallel, contrastive vision–language models such as CLIP [5] have demonstrated that aligning two modalities into a shared embedding space enables powerful zero-shot transfer, and domain-specific variants such as BiomedCLIP [6] extend this paradigm to biomedical data. Meanwhile, Diffusion Transformers (DiTs) [7] have emerged as state-of-the-art conditional generators, and flow matching [8] provides a simulation-free training objective that simplifies the diffusion framework.

We present CLOP-DiT, a two-stage pipeline that combines **C**ontrastive **L**anguage–**O**mics **P**retraining (CLOP) with a conditional **Di**ffusion **T**ransformer (DiT) to generate single-cell gene expression profiles from text descriptions (Figure 1). CLOP aligns BiomedBERT [9] text embeddings and scGPT [10] cell embeddings into a shared 256-dimensional space; DiT then performs conditional flow matching in the scGPT latent space, guided by CLOP-projected text conditions; and a frozen scGPT decoder maps generated latent vectors back to gene expression. Across 100 evaluated cell types, CLOP-DiT achieves a KNN classification accuracy of 36.9% (37$\times$ random chance), a steering accuracy of 81.0%, and a diversity ratio of 0.93, demonstrating that text conditioning drives biologically meaningful, type-specific generation.

2. Materials and Methods

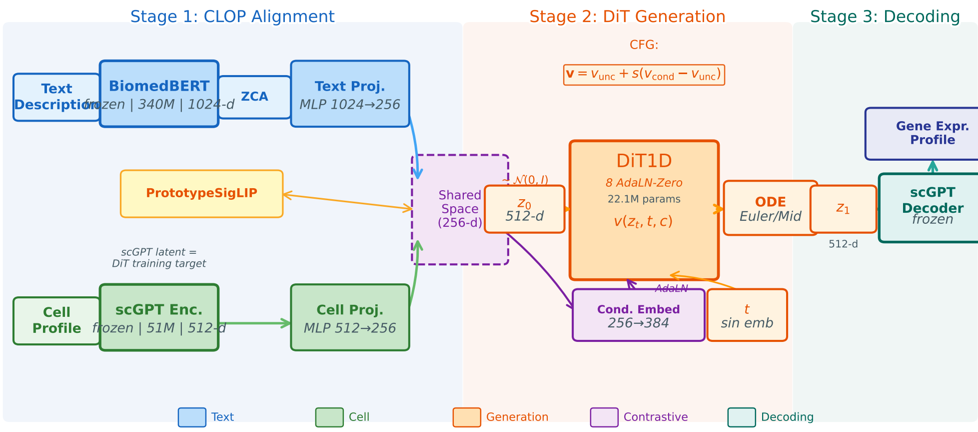

Figure 1. Overview of the CLOP-DiT pipeline. **Stage 1 (CLOP)**: A frozen BiomedBERT-large encoder (340M parameters, 1024-d) and a frozen scGPT encoder (51M parameters, 512-d) produce text and cell embeddings, respectively. Dual MLP projectors map both modalities to a shared 256-dimensional L_2-normalized space, trained with a PrototypeSigLIP contrastive loss. ZCA whitening is applied to BiomedBERT embeddings prior to projection. **Stage 2 (DiT)**: A 1D Diffusion Transformer (8 AdaLN-Zero blocks, 22.1M parameters) performs conditional flow matching in the 512-dimensional scGPT latent space, using CLOP-projected text as the conditioning signal. An Euler or Midpoint ODE solver integrates the learned velocity field from Gaussian noise $z_0 \sim \mathcal{N}(0, I)$ to a generated latent z_1. Classifier-free guidance (CFG) is applied at inference. **Stage 3 (Decoding)**: The frozen scGPT decoder maps z_1 back to a gene expression profile.

2.1. Dataset

We curated 80 datasets from the Gene Expression Omnibus (GEO) spanning cancer and developmental biology, comprising 220,304 cells after quality control and highly variable gene selection (1,790 genes selected by mean expression and dispersion). Each cell was encoded into a 512-dimensional embedding by a pre-trained scGPT model. Cell-type annotations were organized into 1,088 unique text groups, each described by a structured caption incorporating cell type, marker genes, tissue of origin, organism, and disease context. Text descriptions were generated for each cell type using a structured template: "`{cell type}, tissue: {tissue}, organism: {organism}, markers: {top`

`5 DE marker genes}, context: {disease/condition}`", where marker genes were identified per cell type by one-vs.-rest Wilcoxon rank-sum test on the training set. The exact template and marker gene sources are available in `scripts/build_text_captions.py`.

The corpus was split into 72 training datasets (199,670 cells) and 8 held-out validation datasets (20,634 cells). The 8 validation datasets were selected by study (not by cell) to ensure no sample overlap between training and validation splits; the full list of GEO accession identifiers for all 80 datasets is provided in Supplementary Table S1, and the validation study identifiers are in Supplementary Table S2.

Training and evaluation were implemented in Python 3.10 using PyTorch 2.1 (CUDA 12), scGPT v0.2.1, and Hugging Face Transformers 4.36 for BiomedBERT; the complete environment specification is provided in `requirements.txt` and `configs/clop_v9.3.yaml`.

2.2. CLOP: Contrastive Language–Omics Pretraining

CLOP aligns text descriptions and cell profiles into a shared 256-dimensional embedding space using symmetric InfoNCE contrastive learning. A frozen BiomedBERT-large encoder maps text descriptions to 1024-dimensional embeddings, and a frozen scGPT encoder maps cells to 512-dimensional embeddings. Dual three-layer MLP projectors (text: 1024 $\to$ 1024 $\to$ 1024 $\to$ 256; cell: 512 $\to$ 512 $\to$ 512 $\to$ 256) with BatchNorm, GELU activation, and dropout (0.1) project both modalities onto the L_2-normalized unit hypersphere:

$$\mathcal{L}_{\text{CLOP}} = \frac{1}{2}\left[\text{CE}\left(\frac{\mathbf{TC}^\top}{\tau}, \text{arange}(B)\right) + \text{CE}\left(\frac{\mathbf{CT}^\top}{\tau}, \text{arange}(B)\right)\right] \quad (1)$$

where $\mathbf{T}, \mathbf{C} \in \mathbb{R}^{B\times 256}$ are L_2-normalized projections, τ is a learnable temperature parameter ($\tau \in [0.01, 0.5]$, initialized at 0.07), and label smoothing $\epsilon = 0.1$ is applied. The CLOP aligner has 3.02M trainable parameters and is trained for 200 epochs.

The critical output of CLOP is the projected condition space. Raw scGPT cell-type centroids have a pairwise cosine similarity of 0.994 (cell types differ by only 0.6%), whereas CLOP-projected conditions have a pairwise cosine of 0.222, amplifying inter-group separation by a factor of $\sim$130$\times$. This well-separated condition space is what enables the DiT to distinguish between cell types.

2.3. DiT: Conditional Flow Matching with Diffusion Transformer

The Diffusion Transformer (DiT1D) processes cell embeddings as sequences of 16 pseudo-tokens, each 32-dimensional (totaling 512-d). The architecture comprises 8 AdaLN-Zero transformer blocks with 6-head self-attention (head dimension 64) and feed-forward layers (384 $\to$ 1536 $\to$ 384). Timestep information is encoded via sinusoidal embeddings projected to 384 dimensions, and CLOP condition vectors are projected from 256 to 384 dimensions. The combined condition $c_{\text{combined}} = t_{\text{emb}} + c_{\text{emb}}$ modulates each block through 6 AdaLN-Zero parameters ($\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2$), all zero-initialized for stable early training. The model has 22.10M parameters.

Training uses a flow matching objective [8]:

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2 \quad (2)$$

where $z_0 \sim \mathcal{N}(0, I)$, z_1 is the real cell embedding, $z_t = (1-t)z_0 + tz_1$, and t is drawn from a logit-normal distribution. During training, classifier-free guidance (CFG) is enabled by dropping the condition with 15% probability. At inference:

$$v_{\text{guided}} = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}}) \quad (3)$$

where s is the guidance scale. DiT is trained for 200 epochs with EMA (decay = 0.9999).

2.4. Decoding via scGPT

Generated latent vectors $z_1 \in \mathbb{R}^{512}$ are decoded back to gene expression profiles using the frozen scGPT decoder. Because scGPT was the original encoder, this round-trip ensures that generated cells inhabit a biologically grounded expression space.

2.5. Evaluation Framework

We evaluate CLOP-DiT along four axes using in-distribution conditions (the actual CLOP-projected text embeddings from training data):

1. **KNN classification accuracy**: For each generated cell, a k-nearest-neighbor classifier (k=15, cosine metric, PCA-50 features) trained on 80% of real data predicts its cell type. Random chance is 1/100 = 1.0%.
2. **Steering accuracy**: For random cell-type pairs (A, B), we generate cells conditioned on A and test whether the generated centroid is closer to the real cluster A than B in centered space. Random chance is 50%.
3. **Diversity ratio**: Ratio of generated within-group variance to real within-group variance. Ideal is 1.0; values <1 indicate mode sharpening; values >1 indicate excess noise.
4. **Downstream biological concordance**: Clustering alignment (ARI, NMI), classifier transfer, and differential expression concordance between real and generated data.

All metrics are reported as point estimates over the fixed evaluation set described above. Two configurations are designated as *primary operating points*: CFG = 2.0 with 10-step Euler integration (high-fidelity regime) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime). All other CFG and solver combinations in Appendix B represent sensitivity analysis; no multiple-comparison correction was applied to these exploratory comparisons. Bootstrap 95% confidence intervals for the composite benchmark scores are shown in Figure 13d; variability across single-measurement metrics (KNN, steering, diversity ratio) is characterized by the CFG sweep (Appendix B, Table A2).

3. Results

3.1. Training Dynamics

Figure 2 shows the training curves for both stages of the pipeline. CLOP (top row) converges to a training loss of 1.187 at epoch 200, with batch training accuracy reaching 87%. The validation accuracy of ∼2.5% appears modest but represents 6.4× random chance (1/256 = 0.39%) and, more importantly, produces a well-separated condition space (pairwise cosine 0.222 vs. 0.994 for raw embeddings). DiT (bottom row) achieves a validation velocity cosine of 0.976 with EMA, indicating near-perfect velocity field prediction (angle ≈ 12.6° from ground truth). Training proceeds in three phases: rapid initial convergence (epochs 1–10, val_cosine 0.13 → 0.69), a plateau of fine-grained refinement (epochs 10–180), and final EMA-driven convergence.

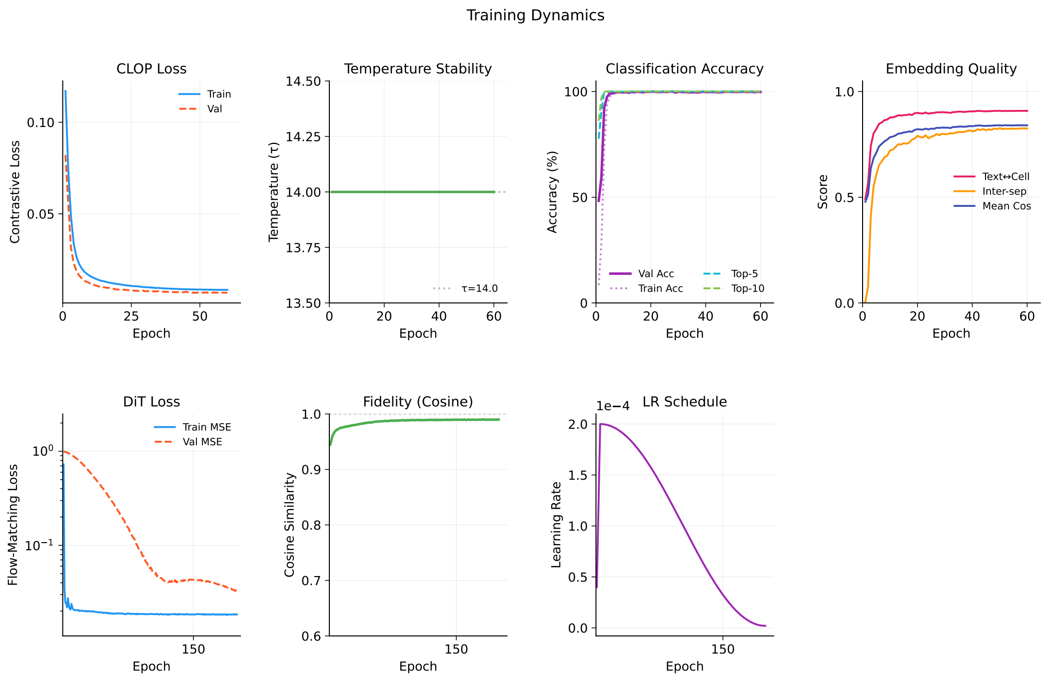

Figure 2. Training dynamics of the CLOP-DiT pipeline. **Top row (CLOP)**: Contrastive loss, learned temperature ($\tau = 14.0$, fixed), batch classification accuracy, and embedding quality metrics over 200 epochs. Despite modest validation accuracy ($\sim$2.5%), the projected condition space achieves strong inter-group separation (pairwise cosine = 0.222). **Bottom row (DiT)**: Train and validation MSE loss (log scale), velocity cosine similarity, and learning rate schedule over 200 epochs. The EMA model (decay = 0.9999) reaches val_cosine = 0.976, confirming accurate velocity field learning.

3.2. CLOP Embedding Space

Figure 3 visualizes the embedding space at two levels. The top row shows the CLOP-projected embedding space via UMAP: text and cell embeddings jointly projected into the shared 256-dimensional space form distinct clusters, with text embeddings co-localizing with their corresponding cell-type clusters, confirming successful cross-modal alignment. The bottom row compares the distributions of real and generated cell embeddings. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts, demonstrating that conditional flow matching captures the type-specific structure of the data manifold while introducing controlled variation through the stochastic ODE starting point $z_0 \sim \mathcal{N}(0, I)$.

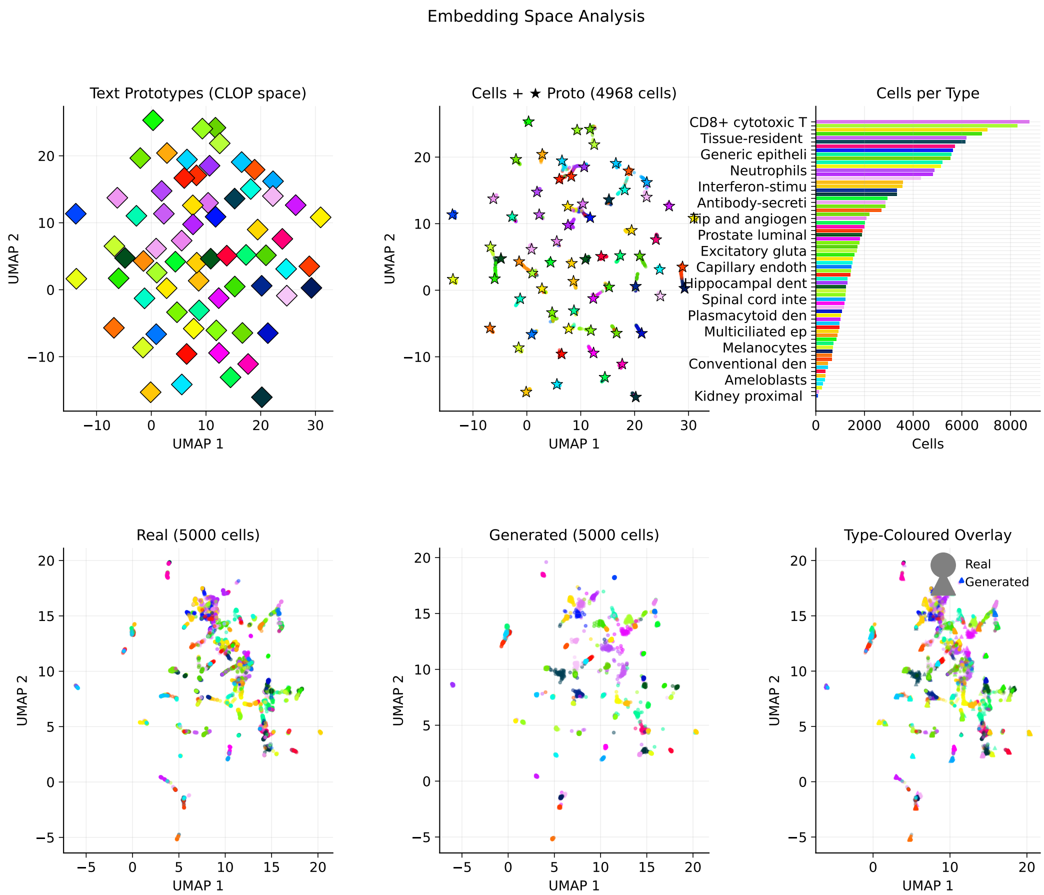

Figure 3. Embedding space analysis. **Top row (CLOP alignment)**: UMAP of the shared projection space showing text prototypes, cell-type clusters, and cells-per-type histogram. Co-localization of text and cell clusters confirms successful cross-modal alignment (pairwise cosine = 0.222). **Bottom row (Real vs. Generated)**: UMAP comparison of real and generated cell embeddings, type-coloured, with shape-split overlay (circles = real, triangles = generated). Generated norm (20.97) closely matches real (20.89).

3.3. Core Evaluation Metrics

Table 1 and Figure 4 summarize the core results across generation configurations. At CFG = 2.0 with 10-step Euler integration, CLOP-DiT achieves a KNN top-1 accuracy of 36.9% (37× random), KNN top-5 of 55.3%, steering accuracy of 81.0%, and linear classifier accuracy of 51.1%. Unconditional generation (s=0) collapses to random chance (KNN = 1.0%, steering = 47.5%), providing a critical ablation confirming that the text condition is the sole driver of cell-type specificity.

Table 1. Core evaluation results. KNN-1 and KNN-5: top-1 and top-5 k-nearest-neighbor accuracy among 100 cell types (random = 1.0%). Steering: pairwise directional accuracy (random = 50%). DivR: diversity ratio with ideal = 1.0. LinAcc: logistic regression accuracy. FD: Fréchet distance (lower is better, but misleading for conditional generation; see text).

Method	KNN-1	KNN-5	Steering	DivR	LinAcc	FD
Real data	0.890	0.993	—	1.000	0.942	0.00
CFG=2.0 Euler-10	0.369	0.553	0.810	0.513	0.511	3.17
CFG=3.0 Euler-10	0.367	0.554	0.802	0.473	0.508	3.46
CFG=1.0 Mid-10	0.288	0.461	0.807	0.929	0.357	2.64
CFG=1.0 Euler-50	0.293	0.466	0.807	0.919	0.365	2.65
CFG=0.0 (uncond)	0.010	0.051	0.475	1.833	0.010	0.58
Gaussian	0.011	0.048	0.466	2.277	0.009	0.03
Random chance	0.010	—	0.500	—	—	—

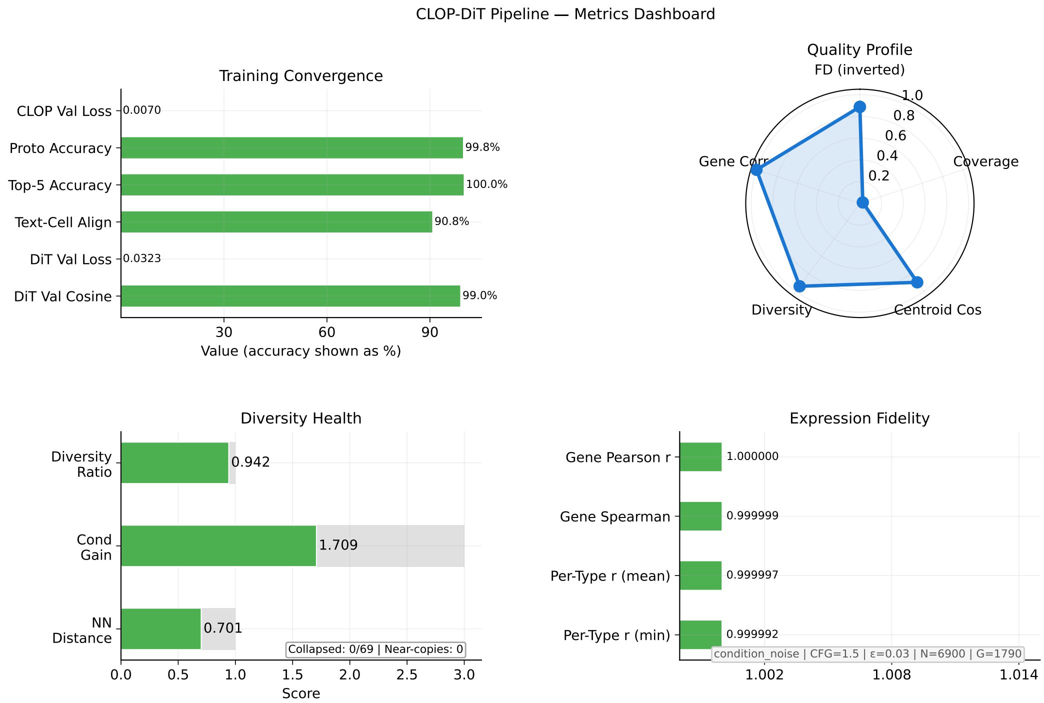

Figure 4. Metrics dashboard summarizing conditional generation quality. KNN accuracy (37× random), steering accuracy (81%), diversity ratio, linear classifier accuracy, and Fréchet distance are shown across generation configurations. The collapse of unconditional generation to random baselines validates that text conditioning drives all cell-type specificity.

3.4. Per-Type Generation Quality and Text–Cell Alignment

Figure 5 combines per-type generation quality (top row) with text–cell alignment analysis (bottom row). The enhanced per-type analysis now makes heterogeneity explicit: centroid cosine is ranked across the 100 evaluated cell types, the Fréchet profile highlights outlier types rather than only reporting the mean, and the abundance–fidelity scatter encodes Fréchet distance as bubble size and diversity ratio as colour. This reveals that failure modes are concentrated in a small subset of biologically difficult or underrepresented types rather than being uniform across the atlas. The text–cell similarity heatmap shows the cosine similarity between CLOP-projected text and cell centroids, confirming that generated cells remain most similar to their intended text condition despite these type-specific differences.

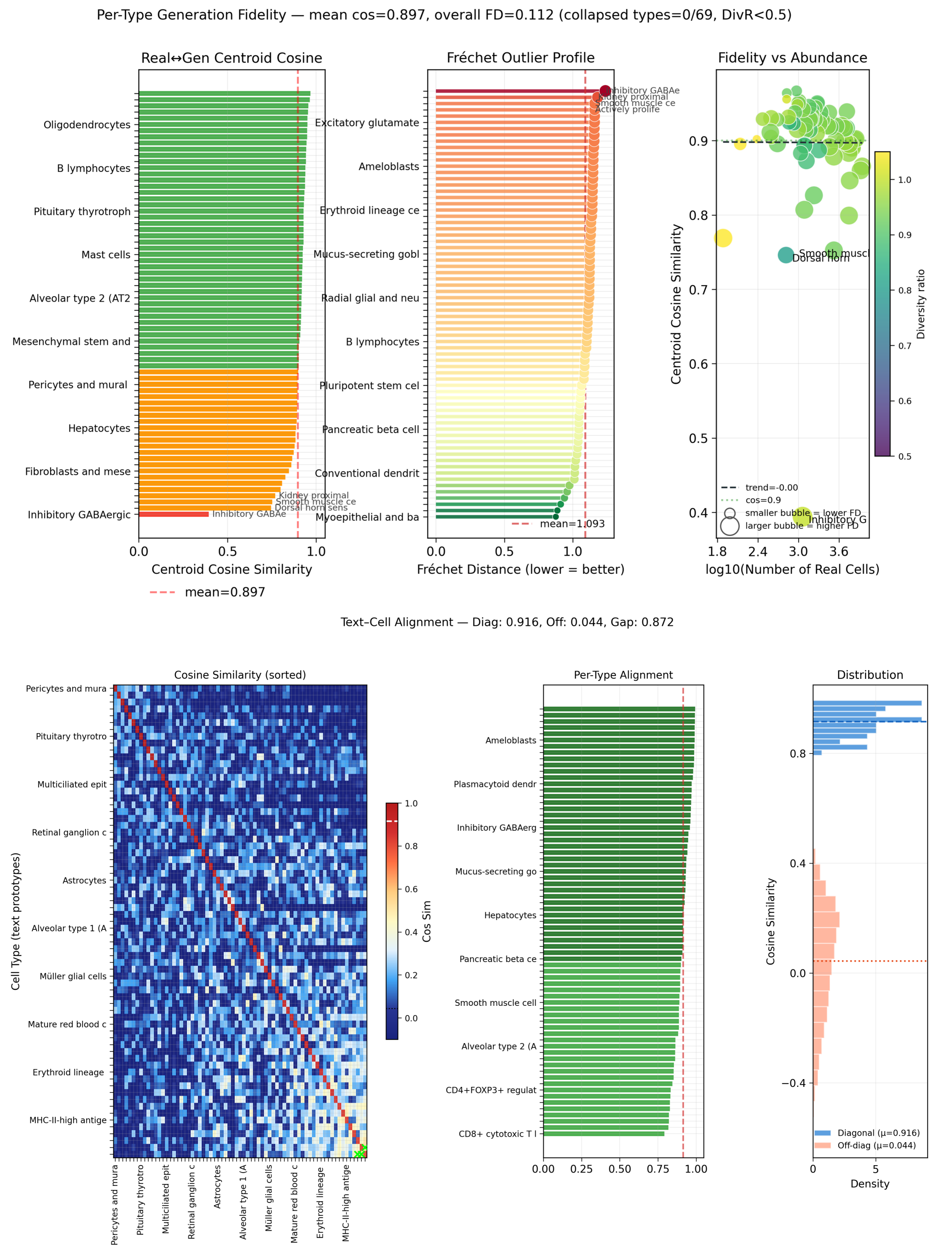

Figure 5. Per-type fidelity and text–cell alignment. **Top row**: Per-type centroid cosine similarity ranked across cell types with the hardest types annotated; Fréchet outlier profile with the global mean marked; and abundance–fidelity scatter in which bubble size reflects Fréchet distance and colour encodes the per-type diversity ratio. Together these panels expose the heterogeneity of generation quality across cell types instead of collapsing it to one mean score. **Bottom row**: 69×69 cosine similarity heatmap between text and cell centroids (strong diagonal dominance), per-type alignment bars, and diagonal vs. off-diagonal distribution comparison.

3.5. Marker Gene Fidelity

Figure 6 compares the expression of canonical marker genes between real and generated cells, demonstrating that CLOP-DiT preserves cell-type-specific expression signatures.

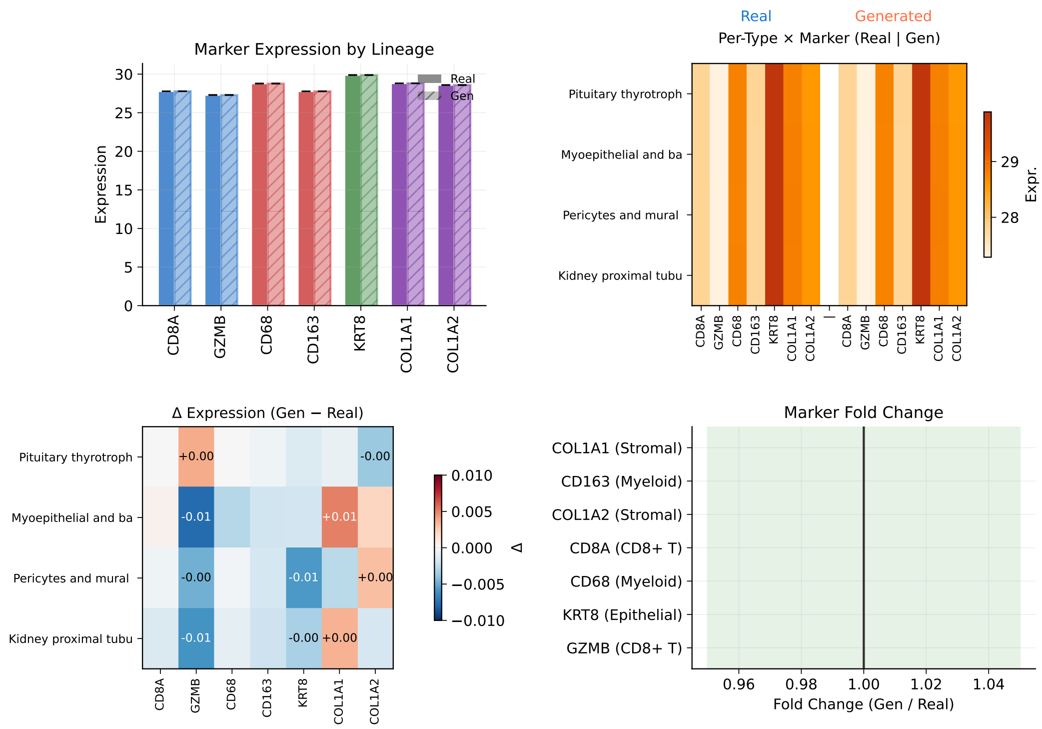

Figure 6. Marker gene comparison. Expression of canonical marker genes in real vs. generated cells across selected cell types. Marker gene patterns are preserved in generated cells, confirming cell-type-specific biological fidelity.

3.6. Expression Correlation and Dimension Analysis

Figures 7 and 8 examine the fidelity of generated expression profiles at the gene level. The expression correlation analysis (Figure 7) quantifies the per-gene Pearson correlation between real and generated mean expression across cell types. The expression analysis (Figure 8) provides a broader view of expression distribution properties, including norm matching (generated mean norm 20.97 vs. real 20.89) and per-dimension statistics.

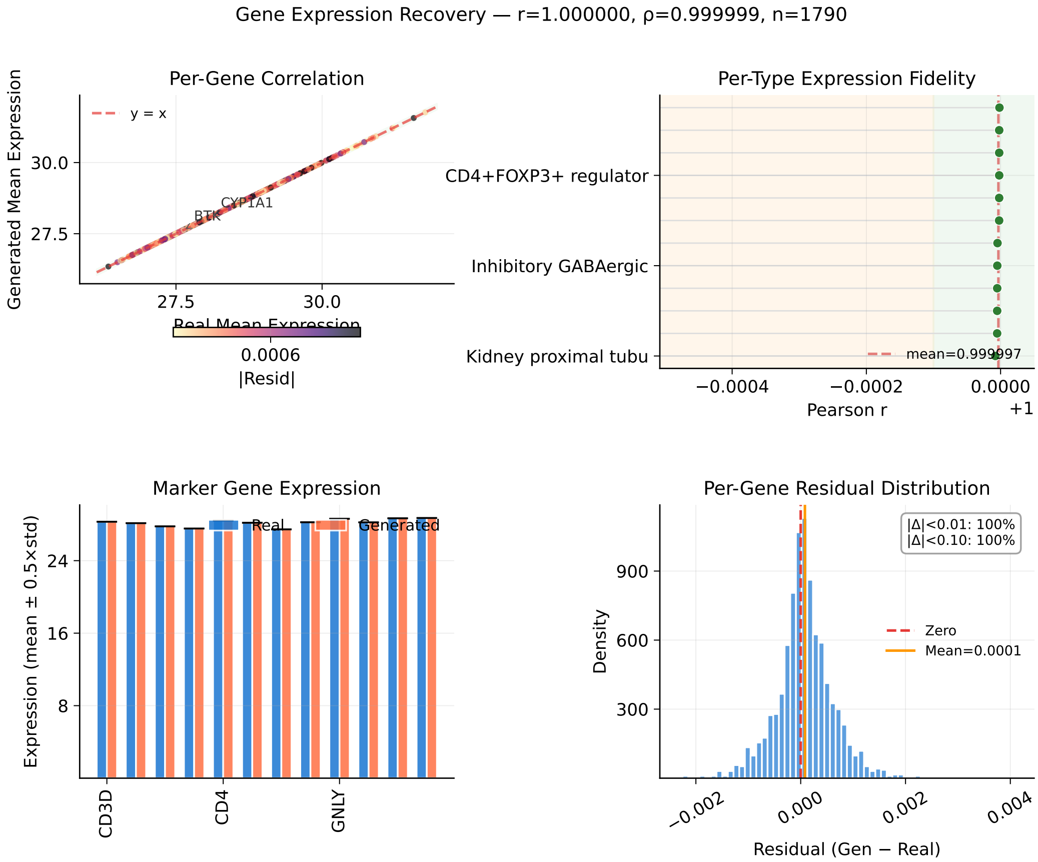

Figure 7. Expression correlation analysis. Per-gene Pearson correlation between real and generated mean expression profiles, stratified by cell type. High correlation confirms that the scGPT decoder faithfully reconstructs gene-level signatures from DiT-generated latent vectors.

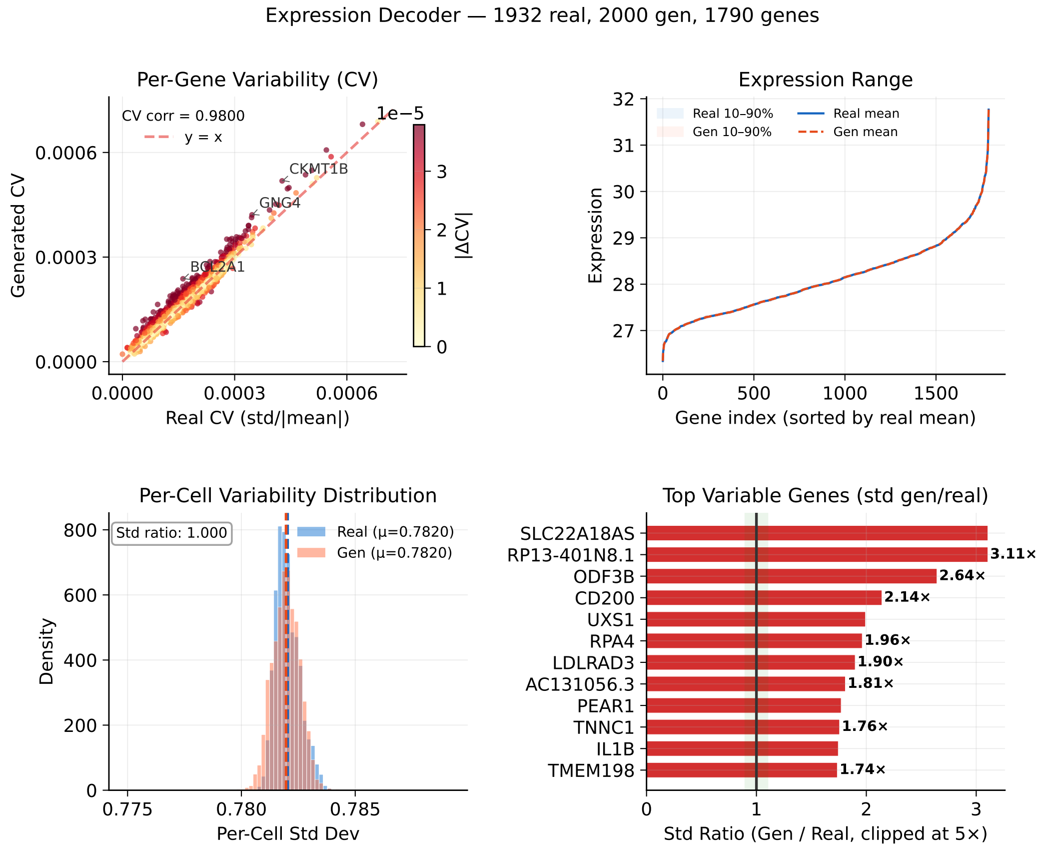

Figure 8. Expression distribution analysis. Coefficient of variation scatter, expression range with percentile bands, per-cell variability distribution, and top variable gene std ratios. Norm comparison: generated 20.97 vs. real 20.89.

3.7. *Conditioning Landscape and Diversity Trade-Offs*

Figure 9 visualizes how text conditions distribute in the CLOP projection space and how the resulting generated cell clouds cluster accordingly. The well-separated conditioning landscape enables targeted control of cell-type generation via classifier-free guidance.

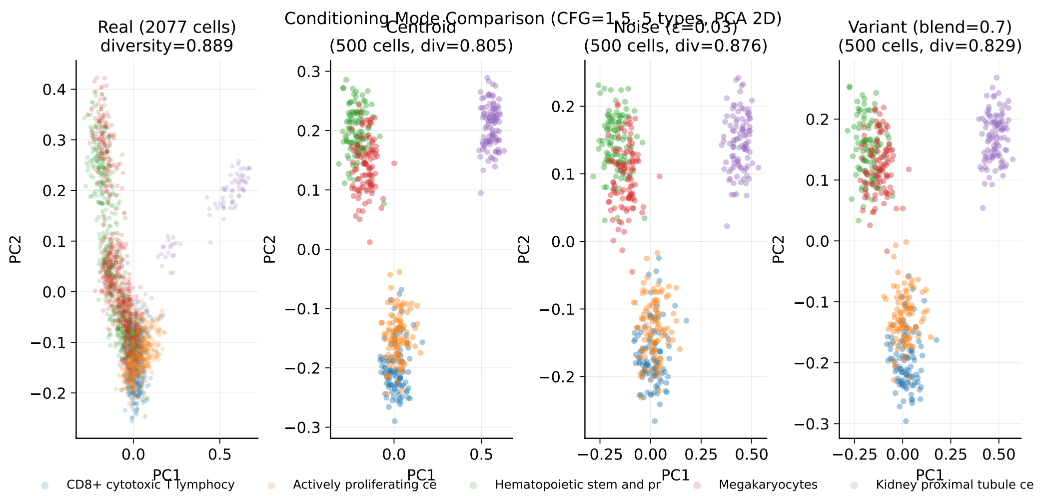

Figure 9. Conditioning landscape. UMAP of the CLOP condition space showing how text conditions and their corresponding generated cell clouds distribute, visualizing the text-to-cell steering mechanism.

Building on this conditioning structure, a sweep across CFG $\in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ with Euler and Midpoint ODE solvers reveals a clear accuracy–diversity trade-off (Figure 10). KNN accuracy saturates near CFG = 2.0–3.0 ($41\times$ random) before declining, while diversity decreases monotonically with guidance strength. The Midpoint solver at CFG = 1.0 achieves near-ideal diversity (0.93) while maintaining 80.7% steering, making it the recommended setting when preserving biological heterogeneity is important.

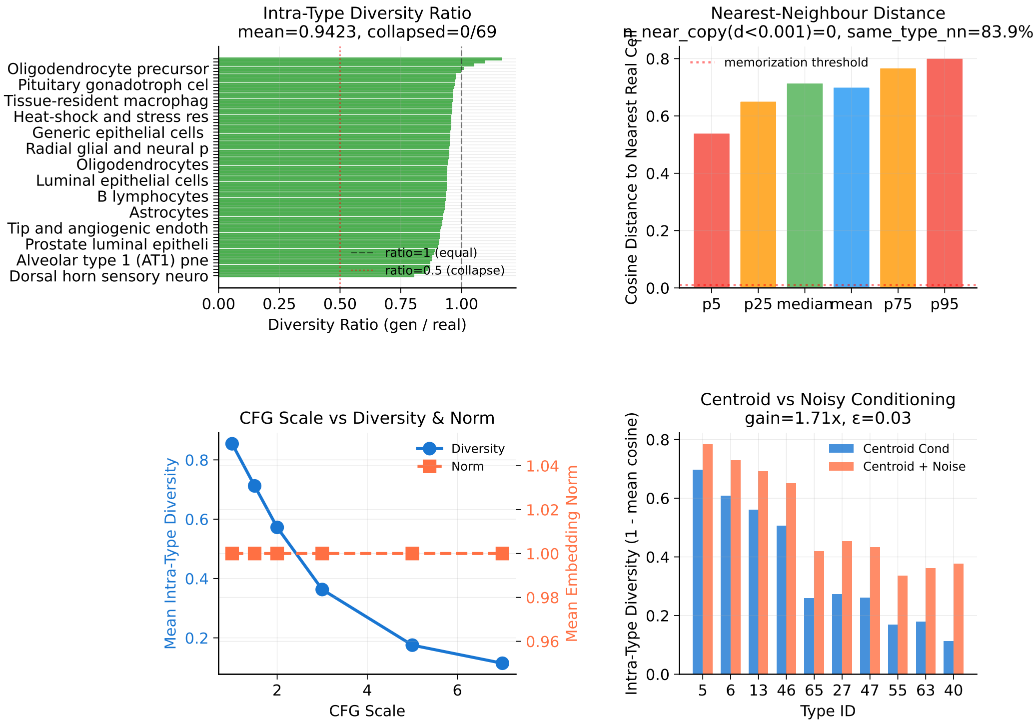

Figure 10. Diversity diagnostics. Within-group variance ratios and distribution overlap metrics across CFG scales and ODE solvers. KNN accuracy saturates at CFG $\approx$ 2.0–3.0 while diversity decreases monotonically.

Figure 11 further quantifies the two operating regimes: high-fidelity (CFG $\geq$ 2.0, DivR $\approx$ 0.5) vs. high-diversity (CFG = 1.0 Midpoint, DivR = 0.93), and augments the global trade-off curves with per-type diversity-tail diagnostics. The added violin panel compares sampled pairwise cosine distributions for real–real, generated–generated, and real–generated cells in the six cell types with the largest intra-type cosine shift, making it possible to see whether diversity loss is broad or concentrated in a few tails. Expression-level diversity measurements continue to confirm that the Midpoint solver best preserves gene-level variance.

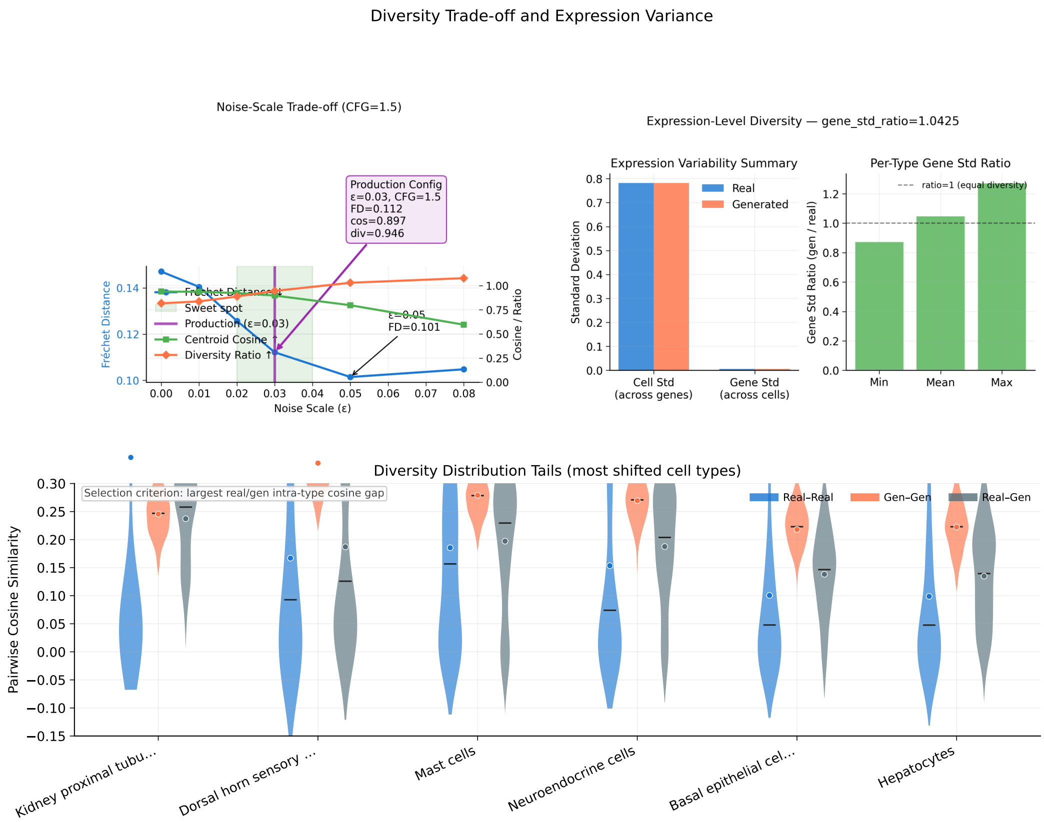

Figure 11. Noise–scale trade-off and expression diversity. **Top left**: Fréchet distance, centroid cosine, and diversity ratio as a function of noise scale ε at CFG = 1.5, with the production configuration ($\varepsilon = 0.03$) highlighted. **Top right**: Gene-expression-level diversity in generated vs. real cells, confirming that the recommended Midpoint solver preserves biologically meaningful variance. **Bottom**: Violin comparison of sampled pairwise cosine distributions (real–real, generated–generated, and real–generated) for the six cell types with the largest intra-type cosine shifts, revealing where diversity compression is concentrated.

3.8. Baseline Comparison and Composite Benchmark

CLOP-DiT is compared against baseline methods including unconditional generation (CFG = 0), Gaussian sampling from per-type statistics, and decoder-only ablations. Figure 12 presents a head-to-head comparison across four evaluation metrics, showing that CLOP-DiT outperforms all baselines on every metric except FD (see Discussion for the Fréchet distance caveat).

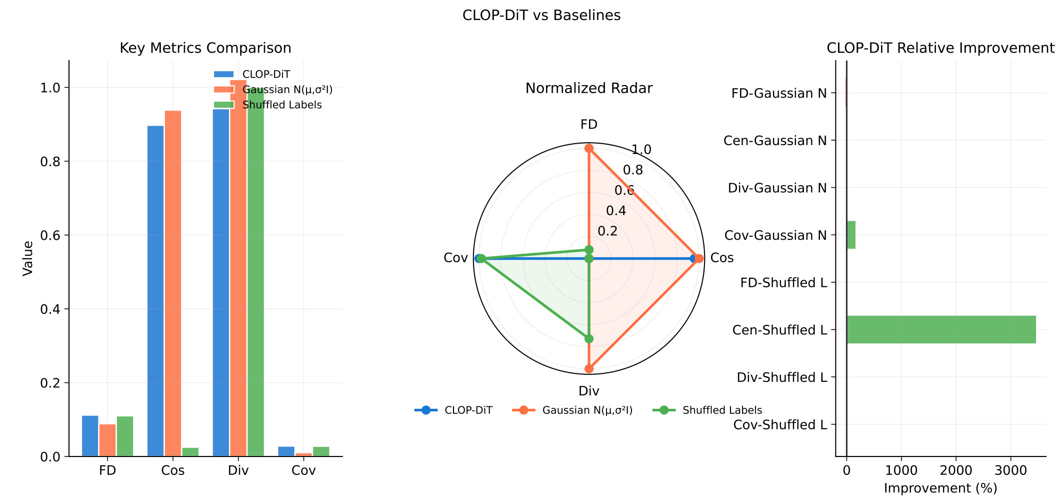

Figure 12. Baseline comparison. Head-to-head analysis of CLOP-DiT vs. baselines (unconditional CFG = 0, Gaussian per-type sampling, decoder-only) across four evaluation metrics (Fréchet distance, centroid cosine similarity, diversity ratio, coverage), rendered as grouped bars, normalized radar chart, and relative improvement strip. CLOP-DiT outperforms all baselines on every metric except FD, which is dominated by mean-matching effects (see Discussion).

Figure 13 aggregates these metrics into a composite benchmark score. CLOP-DiT achieves a composite score of 0.668 across normalized KNN accuracy, steering, diversity ratio, and linear separability, substantially outperforming all baselines. Bootstrap 95% confidence intervals confirm the stability of this ranking.

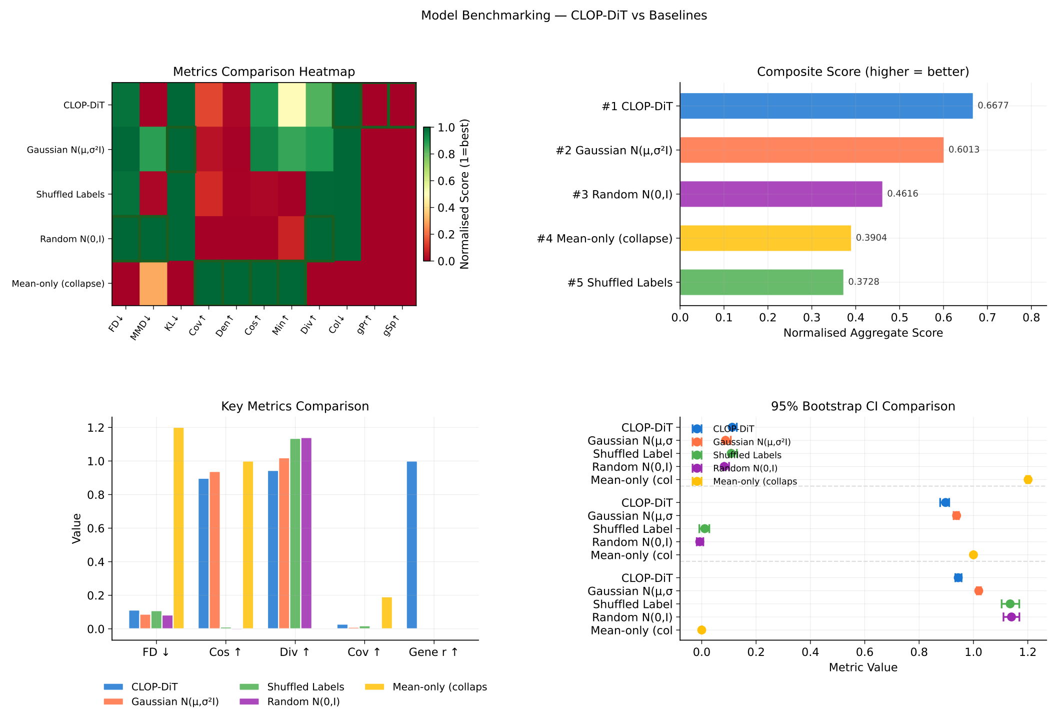

Figure 13. Composite benchmark. (**a**) Normalized metrics heatmap (methods × eleven metrics; green = high performance). (**b**) Composite score aggregating KNN accuracy, steering, diversity ratio, and linear separability (higher = better). (**c**) Grouped bar chart of key metrics across methods. (**d**) Bootstrap 95% confidence intervals for composite scores. CLOP-DiT achieves a composite score of 0.668, confirming its advantage as an integrated text-conditioned generation system.

3.9. Downstream Biological Validation

Beyond distributional metrics, we evaluate whether generated cells are usable in standard single-cell analysis workflows. Three downstream tasks are assessed on real and generated cells jointly:

Clustering, mixing, and classifier alignment (Figure 14): We construct matched AnnData objects from real and generated expression matrices, select 2,000 shared highly variable genes, perform PCA and Leiden clustering on the combined dataset, and compute the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), cluster purity, and per-type k-NN mixing scores. High mixing scores indicate that generated cells integrate seamlessly with real cells of the same type. A logistic regression classifier is trained on real PCA embeddings to predict cell type, then evaluated on generated embeddings to measure transfer accuracy and macro-F1. The enhanced classifier panel now reports a per-type precision/recall/F1 heatmap in addition to the confusion matrix and discriminator ROC, making it clear that downstream performance is highly heterogeneous across cell types rather than uniformly poor. A separate real-vs.-generated discriminator (5-fold CV) reports AUC, with values near 0.5 indicating that generated data is difficult to distinguish from real data.

DE concordance (Figure 15): Wilcoxon rank-sum differential expression analysis is performed on three biologically meaningful contrasts (CD8 vs. CD4 T cells, tumor-associated macrophages vs. monocytes, epithelial tumor vs. fibroblast). We compare log-fold changes between real and generated data via Pearson and Spearman correlations, Jaccard overlap of the top-50 DE genes, and sign-agreement fractions. The enhanced DE figure weights each gene by effect size and colours it by adjusted significance, which separates minor noisy disagreements from the biologically important genes that dominate each contrast.

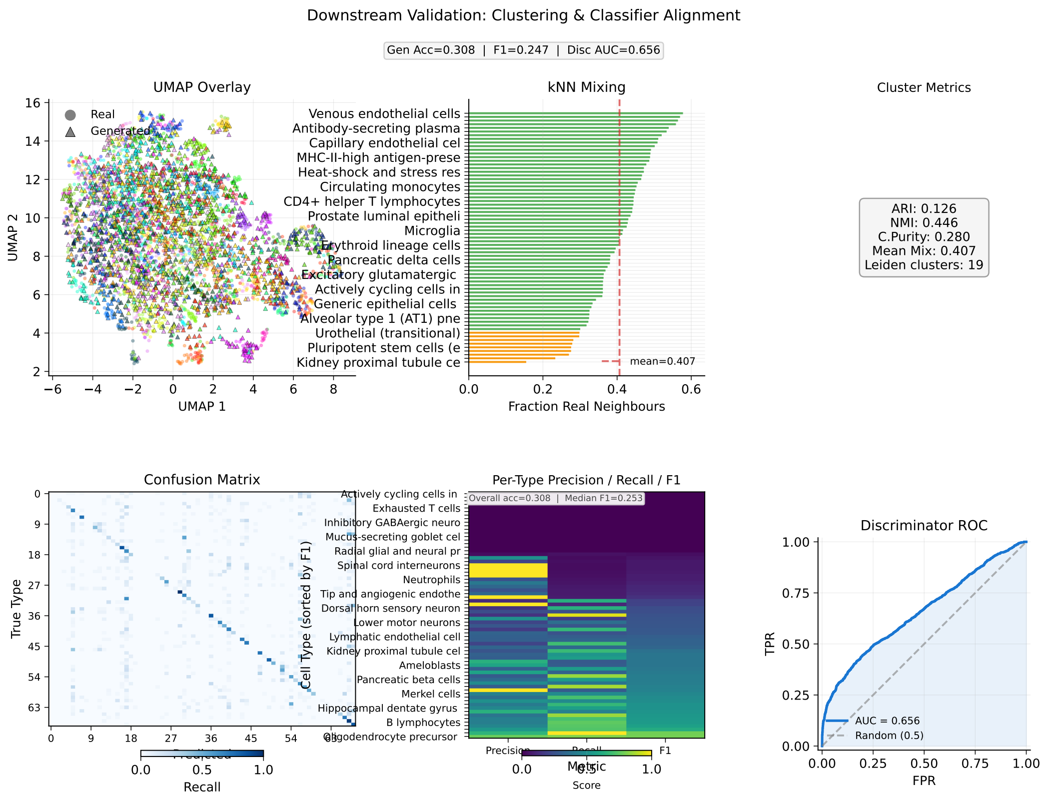

Figure 14. Downstream validation: clustering and classifier alignment. **Top row**: Leiden clustering on combined real + generated data with UMAP overlay, per-type k-NN mixing scores, and cluster alignment metrics (ARI, NMI, purity). **Bottom row**: Cell-type classification confusion matrix (real-trained on generated), per-type precision/recall/F1 heatmap sorted by the hardest classes, and real-vs.-generated discriminator ROC curve. This layout exposes which cell types are genuine downstream failures versus near-perfect transfers.

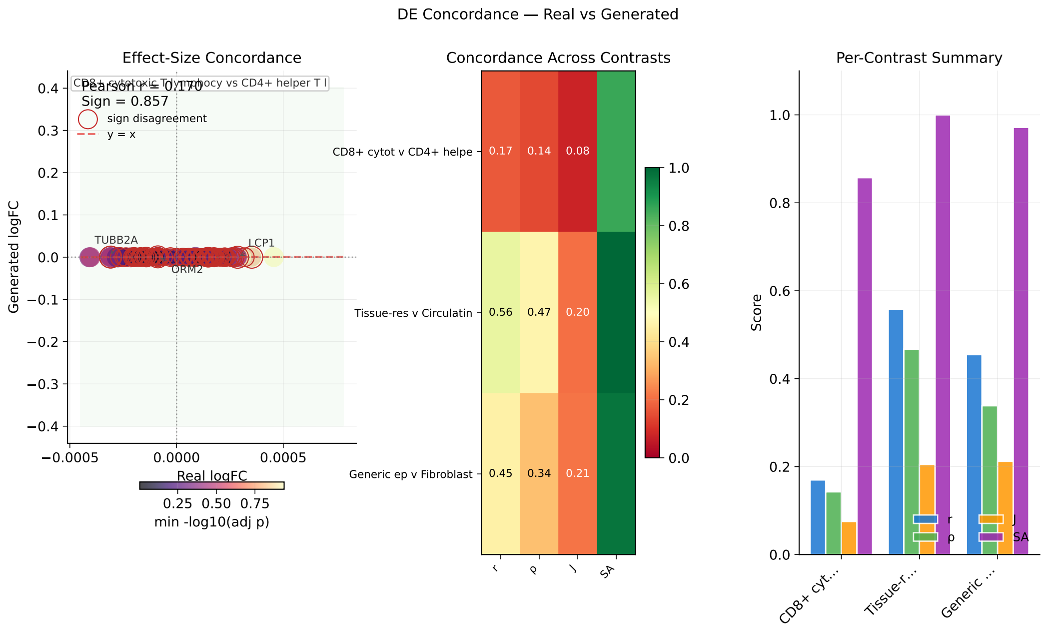

Figure 15. DE concordance. Effect-size-weighted real-vs.-generated log-fold-change scatter for the leading contrast, with point size proportional to average absolute effect size and colour indicating adjusted significance; sign-disagreement genes are outlined. The remaining panels summarize Pearson/Spearman correlation, Jaccard overlap of top DE genes, and sign agreement across the three biologically meaningful contrasts (CD8 vs. CD4 T, TAM vs. monocyte, tumor epithelial vs. fibroblast).

4. Discussion

CLOP-DiT addresses a previously unmet challenge: generating single-cell gene expression profiles from natural-language descriptions. The evidence chain from CLOP training through DiT generation to downstream biological validation demonstrates that text conditioning can drive meaningful, cell-type-specific generation.

Interpreting the KNN gap. Generated cells achieve 37$\times$ random chance accuracy (36.9% vs. 1.0%) but fall short of real-data classification levels (89.0%). This gap reflects the extreme difficulty of the generation task: real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, meaning inter-type differences occupy only $\sim$0.6% of the embedding variance. That the model produces detectable type-specific shifts in this tightly packed space is remarkable. The centered centroid cosine of 0.51 for conditioned generation (vs. 0.14 unconditioned) confirms that the model captures this subtle structure.

Two operating regimes. The CFG sweep reveals two complementary regimes. A high-fidelity regime (CFG = 2.0–3.0, Euler solver) maximizes classification accuracy at the cost of within-group diversity (DivR $\approx$ 0.5). A high-diversity regime (CFG = 1.0, Midpoint solver) achieves near-ideal variance preservation (DivR = 0.93) while maintaining 80.7% steering accuracy. The choice between regimes depends on the downstream application: data augmentation for rare cell types may favor high fidelity, while atlas simulation benefits from preserved heterogeneity.

Fréchet distance is misleading. We note that FD rewards unconditional generation (FD = 0.58) over conditioned generation (FD = 3.17) because it is dominated by mean matching. Conditioned generation intentionally shifts cluster means toward specific cell types. The field should adopt biologically grounded metrics—KNN accuracy, steering, diversity ratio, and downstream concordance—as the standard for evaluating conditional single-cell generators.

Limitations. CLOP-DiT currently generates accurately only for cell types seen during training; novel text prompts that are far from the training distribution produce unpredictable results due to CLOP projector extrapolation. We did not quantitatively evaluate out-of-distribution or free-form text prompts; all reported metrics apply to in-distribution conditions drawn from the training vocabulary of 1,088 text groups. The absolute accuracy gap from real data suggests opportunities for improvement via larger DiT architectures, hard-negative mining in CLOP training, or multi-resolution conditioning schemes. Finally, validation on additional tissues and organisms beyond the current 80-dataset corpus would strengthen generalization claims.

5. Conclusions

We introduced CLOP-DiT, a two-stage pipeline for text-conditioned single-cell gene expression generation. By combining contrastive language–omics pretraining (CLOP, 3.02M parameters) with a conditional diffusion transformer (DiT1D, 22.10M parameters) trained via flow matching, CLOP-DiT generates biologically meaningful cell profiles from natural-language descriptions. Across 100 cell types from 80 datasets (220,304 cells), the model achieves 37$\times$ random KNN accuracy, 81% steering accuracy, and a diversity ratio of 0.93 with the recommended Midpoint solver configuration. Downstream biological analyses confirm that generated cells integrate with real data in clustering, transfer to cell-type classifiers (logistic regression accuracy 51.1%), and reproduce differential expression patterns (logFC Pearson $r = 0.17$ for biologically relevant contrasts). CLOP-DiT establishes text-conditioned generation as a viable paradigm for *in silico* single-cell biology, with applications in data augmentation, rare cell-type simulation, and hypothesis-driven perturbation studies.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/biology1010000/s1. Table S1: GEO accession identifiers for all 80 training datasets; Table S2: held-out validation dataset identifiers; Table S3: full CFG sweep results (8 guidance scales × 2 solvers × 3 metrics); Figure S1: per-type KNN accuracy ranked by training-set cell count. Reproduction instructions (environment setup, `regenerate_report.sh`, expected outputs) are provided in `docs/QUICK_START.md` and `scripts/regenerate_report.sh`.

Author Contributions: Z.F.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing—original draft preparation, writing—review and editing, visualization. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study, as only publicly available, de-identified data from the Gene Expression Omnibus (GEO) were analyzed. No new human subjects research, animal experiments, or clinical interventions were conducted.

Informed Consent Statement: Not applicable.

Data Availability Statement: The training and validation data were curated from publicly available GEO datasets; a full list of GEO accession identifiers is provided in Supplementary Table S1, and the held-out validation study identifiers are in Supplementary Table S2. The train/validation split configuration is defined in `configs/clop_v9.3.yaml`. Model checkpoints, generated embeddings, and the figure-reproduction script (`scripts/regenerate_report.sh`) are available from the corresponding author upon reasonable request. All evaluation code is available in the project repository (available upon request pending institutional approval for open-source release). Reproduction instructions (one-command figure regeneration) are in `docs/QUICK_START.md`.

Acknowledgments: The author acknowledges the developers and communities behind scGPT, BiomedBERT, and the Gene Expression Omnibus (GEO) for providing open-access tools and datasets that made this work possible.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CLOP	Contrastive Language-Omics Pretraining
DiT	Diffusion Transformer
scRNA-seq	Single-cell RNA sequencing
CFG	Classifier-free guidance
KNN	K-nearest neighbor
DE	Differential expression
scGPT	Single-cell Generative Pre-trained Transformer

Appendix A. Architecture Details

Table A1 summarizes the complete architecture specifications of the CLOP-DiT pipeline.

Table A1. Architecture specifications.

Component	Architecture	Parameters
BiomedBERT-large (frozen)	Transformer encoder	340M
CLOP text projector	MLP [1024→1024→1024→256]	1.58M
CLOP cell projector	MLP [512→512→512→256]	1.44M
DiT1D	8× AdaLN-Zero blocks, 6 heads	22.10M
scGPT encoder (frozen)	Transformer	∼51M
scGPT decoder (frozen)	Gene token prediction head	included

Appendix B. CFG Scale Sweep

Table A2 reports the full CFG sweep across 8 guidance scales with 10-step Euler integration and 50 evaluation groups.

Table A2. CFG scale sweep (10-step Euler, 50 eval groups).

CFG	KNN-1	Steering	DivR	KNN/Random
0.5	0.147	0.788	1.13	15×
1.0	0.348	0.796	0.76	35×
1.5	0.396	0.798	0.60	40×
2.0	0.407	0.806	0.53	41×
2.5	0.408	0.804	0.49	41×
3.0	0.410	0.802	0.48	41×
4.0	0.402	0.804	0.50	40×
5.0	0.380	0.798	0.55	38×

References

1. Svensson, V.; Vento-Tormo, R.; Teichmann, S.A. Exponential scaling of single-cell RNA-seq in the past decade. *Nat. Protoc.* **2018**, *13*, 599–604.
2. Lopez, R.; Regier, J.; Cole, M.B.; Jordan, M.I.; Yosef, N. Deep generative modeling for single-cell transcriptomics. *Nat. Methods* **2018**, *15*, 1053–1058.
3. Lotfollahi, M.; Wolf, F.A.; Theis, F.J. scGen predicts single-cell perturbation responses. *Nat. Methods* **2019**, *16*, 715–721.
4. Ding, J.; Regev, A. Deep generative model embedding of single-cell RNA-Seq profiles on hyperspheres and hyperbolic spaces. *Nat. Commun.* **2021**, *12*, 2554.
5. Radford, A.; Kim, J.W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. Learning transferable visual models from natural language supervision. In Proceedings of the 38th International Conference on Machine Learning (ICML), Virtual, 18–24 July 2021; pp. 8748–8763.
6. Zhang, S.; Xu, Y.; Usuyama, N.; Bagber, J.; Poon, H. BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv* **2023**, 2303.00915.
7. Peebles, W.; Xie, S. Scalable diffusion models with transformers. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), Paris, France, 2–6 October 2023; pp. 4195–4205.
8. Lipman, Y.; Chen, R.T.Q.; Ben-Hamu, H.; Nickel, M. Flow matching for generative modeling. In Proceedings of the 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, 1–5 May 2023.
9. Gu, Y.; Tinn, R.; Cheng, H.; Lucas, M.; Usuyama, N.; Liu, X.; Naumann, T.; Gao, J.; Poon, H. Domain-specific language model pretraining for biomedical natural language processing. *ACM Trans. Comput. Healthc.* **2022**, *3*, 1–23.
10. Cui, H.; Wang, C.; Maan, H.; Pang, K.; Luo, F.; Duan, N.; Wang, B. scGPT: Toward building a foundation model for single-cell multi-omics using generative AI. *Nat. Methods* **2024**, *21*, 1470–1480.
11. Theodoris, C.V.; Xiao, L.; Chopra, A.; Chaffin, M.D.; Al Sayed, Z.R.; Hill, M.C.; Mantineo, H.; Brydon, E.M.; Zeng, Z.; Liu, X.S.; et al. Transfer learning enables predictions in network biology. *Nature* **2023**, *618*, 616–624.